

CCNA 200-301 V2.0

11

EtherChannel and LACP

Part II — Switching and Network Access

EXAM TOPICS

2.1

LECTURER

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What you will be able to do

- **Explain** why bundling links needs a protocol rather than just more cables, in terms of what spanning tree would otherwise do.
- **Configure** a layer 2 EtherChannel with LACP, and a layer 3 EtherChannel between routed ports.
- **Verify** a bundle and read the flags in `show etherchannel summary`.
- **Diagnose** the parameter mismatches that stop a bundle forming — the single most common EtherChannel fault.

Before we start

Chapter 10 for trunks and allowed VLAN lists; Chapter 8 for access ports.

Vocabulary for this chapter

EtherChannel

port channel

LACP

PAgP

active

passive

on

load balancing

bundle

suspended port

layer 3 EtherChannel

Each is defined where it first matters — not here.

EtherChannel

An **EtherChannel** bundles two to eight physical links into one logical interface, the **port channel**. Spanning tree sees a single link and does not block any member, so all of the bandwidth is usable, and if one member fails the others carry on without a topology change.

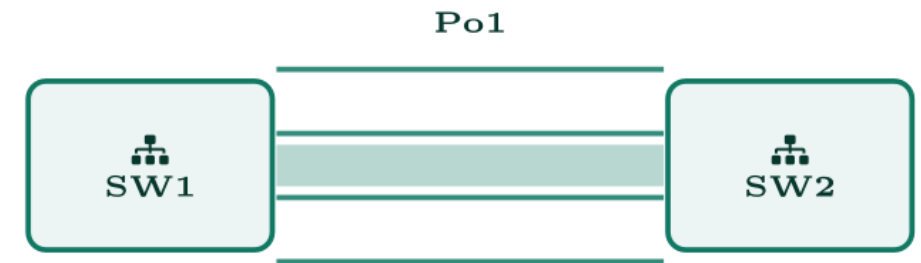
Why bundling needs a protocol

Four cables, no EtherChannel



spanning tree blocks three.
You paid for 4 Gbps and use 1

Four cables, one port channel



STP sees *one* link, so none is blocked.
All 4 Gbps is usable

Bundling is not about speed for one conversation — it is about spanning tree seeing one link instead of four, and therefore blocking none of them.

Load balancing is per conversation, not per packet

Frames are distributed across members by hashing fields from the frame — source MAC, destination MAC, source and destination IP, or a combination. Every frame in one conversation hashes the same way, so a conversation always takes one member link.

The consequence catches people out: a single large file transfer between two hosts gets *one member's worth* of bandwidth, not the whole bundle. Four 1 Gbps links do not give one server a 4 Gbps transfer. They give you 4 Gbps of aggregate capacity across many conversations, which is a different thing. When somebody complains that a new EtherChannel “did not make the backup faster”, this is why.

Never mix on with a negotiating mode

`on` at one end and `active` at the other does not form a bundle, and worse, the `on` end starts forwarding on every member immediately while the far end still treats them as individual links. That is a loop, built by hand. Either both ends negotiate, or both ends are `on`. LACP `active` at both ends is the right default.

The rule that governs everything: members must match

Every member port in a bundle, *at both ends*, must agree on:

- speed and duplex;
- switch mode — all access, or all trunk;
- if trunking: the same native VLAN and the same allowed VLAN list;
- if access: the same access VLAN.

One port that disagrees is suspended and stays out of the bundle. Configure the port channel interface and let the settings inherit downward, rather than configuring each member separately — that is how you avoid the mismatch in the first place.

A two-member lacp trunk bundle, both ends

Configure

```
! Put the physical members into the channel group first. Creating  
! the group creates the Port-channel interface automatically.  
interface range GigabitEthernet0/1 - 2  
    description Member of Po1 - uplink to SW2  
    shutdown  
    channel-group 1 mode active  
    no shutdown  
exit  
!  
! Now configure the logical interface. Settings applied here  
! propagate to every member, which is what keeps them matching.  
interface Port-channel1  
    description Uplink bundle to SW2  
    switchport trunk encapsulation dot1q  
    switchport mode trunk  
    switchport trunk native vlan 999  
    switchport trunk allowed vlan 10,20,30  
!  
! Hash on both source and destination IP for a better spread than  
! the default source-MAC-only on many platforms.  
port-channel load-balance src-dst-ip
```

show etherchannel summary

```
SW1# show etherchannel summary
```

```
Flags:  D - down          P - bundled in port-channel
        I - stand-alone  s - suspended
        H - Hot-standby (LACP only)
        R - Layer3       S - Layer2
        U - in use       f - failed to allocate aggregator
        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port
```

```
Number of channel-groups in use: 1
```

```
Number of aggregators: 1
```

Group	Port-channel	Protocol	Ports
1	Po1(SU)	LACP	Gi0/1(P) Gi0/2(P)

A routed bundle

Configure

```
interface Port-channel2
  description Routed bundle to Core2
  ! no switchport makes this a routed interface -- do it BEFORE
  ! adding an address, or the address command is rejected.
  no switchport
  ip address 10.255.0.1 255.255.255.252
  !
interface range GigabitEthernet0/23 - 24
  no switchport
  channel-group 2 mode active
  no shutdown
```

A layer 3 EtherChannel

- **Passive–passive or auto–auto.** Neither end initiates and no bundle forms. The ports stay up as individual links, which spanning tree then blocks — so you get a working network and half the bandwidth you paid for.
- **Mismatched allowed VLAN lists.** A favourite: the bundle forms, one VLAN is missing from one end, and only that VLAN breaks.
- **Configuring members individually.** Configure `interface Port-channel1`; let it inherit.
- **Adding no `switchport` after the IP address.** On a layer 3 bundle the order matters.
- **Expecting one flow to use the whole bundle.** It never will.

The bundle formed with one member instead of four.

A new four-link bundle shows only `Gi0/1` bundled; the other three are suspended.

The bundle formed with one member instead of four.

What would you check first — and what do you expect to see?

show etherchannel summary | begin Group

```
SW1# show etherchannel summary | begin Group
```

Group	Port-channel	Protocol	Ports
1	Po1(SU)	LACP	Gi0/1(P) Gi0/2(s) Gi0/3(s) Gi0/4(s)

show interfaces GigabitEthernet0/2 switchport | include Access|Trunking|Mode

```
SW1# show interfaces GigabitEthernet0/2 switchport | include Access|Trunking|Mode
```

```
Administrative Mode: static access  
Operational Mode: static access  
Access Mode VLAN: 1 (default)  
Trunking Native Mode VLAN: 1 (default)
```

What is the fault — and what does this output rule out?

Why it is doing that

Diagnosis. `Gi0/1` was configured as a trunk before being added to the group, and the others were left as access ports in VLAN 1. Members must agree on switch mode, so the three that disagree with the first are suspended. The `(s)` flag is the whole diagnosis; the second command identifies which parameter differs.

And the lesson

Fix. Do not fix the members one by one — that invites the same fault again. Apply the trunk configuration to `interface Port-channel1` and let it propagate, then confirm all four show (P). If a member still refuses, compare speed and duplex next.

Bundle four links and break them

Topology. Two switches joined by four Ethernet links; one PC on each.

1. Cable all four links with no EtherChannel configured. Run `show spanning-tree` and record how many are forwarding and how many are blocking. Explain the result.
2. Configure `Po1` with LACP `active` at both ends, trunked, native VLAN 999, allowed 10 and 20. Verify with `show etherchannel summary`.
3. Re-run `show spanning-tree`. How many logical links does it now see?
4. Shut one member. Confirm traffic continues and that spanning tree does not reconverge.
5. Set one member's access VLAN to 20 by configuring the physical port directly. Record the flag it earns and explain it.
6. Change one end to `passive` while the other is also `passive`. Record what the bundle does.

Bundle four links and break them (continued)

1. Extension. Build a layer 3 EtherChannel between two multilayer switches with a /30 and prove you can ping across it.

CHECKPOINT 1 of 3

Two switches are configured channel-group 1 mode passive at both ends. What happens, and what does spanning tree do about it?

Discuss · then advance

Checkpoint 1 of 3

Two switches are configured channel-group 1 mode passive at both ends. What happens, and what does spanning tree do about it?

No channel forms — both ends wait to be asked. The individual links stay up, so spanning tree sees parallel paths and blocks all but one. The result is a working but single-link connection, which is why the symptom is “it works, but it is not four gigabits”.

CHECKPOINT 2 of 3

A member port shows (s). Name three parameters you would compare and the command that shows them.

Discuss · then advance

Checkpoint 2 of 3

A member port shows (s). Name three parameters you would compare and the command that shows them.

(s) is suspended: a parameter mismatch. Compare speed, duplex, and the allowed VLAN list and native VLAN on each member. `show interfaces <int> switchport` and `show interfaces status` show them.

CHECKPOINT 3 of 3

Why does adding a second member to a bundle not double the speed of a single file copy?

Discuss · then advance

Checkpoint 3 of 3

Why does adding a second member to a bundle not double the speed of a single file copy?

Because load balancing is per-*flow*, not per-packet. A hash of addresses and ports selects one member for a conversation, and one file copy is one conversation. Bundling raises aggregate capacity across many flows, not the speed of any single one.

What to take away

- An EtherChannel makes several links look like one, so spanning tree blocks none of them.
- LACP **active** at both ends is the sensible default. Passive–passive and auto–auto form nothing; **on** mixed with a negotiating mode builds a loop.
- Members must match on speed, duplex, switch mode, native VLAN and allowed list. Configure the port channel interface and let it inherit.
- Read `show etherchannel summary` by its flags: **(P)** bundled, **(s)** suspended for a mismatch, **(I)** stand-alone, **(D)** down.
- Load balancing is per conversation. One flow uses one member.

Where we got to

- An EtherChannel makes several links look like one, so spanning tree blocks none of them.
- LACP active at both ends is the sensible default. Passive–passive and auto–auto form nothing; on mixed with a negotiating mode builds a loop.
- Members must match on speed, duplex, switch mode, native VLAN and allowed list. Configure the port channel interface and let it inherit.
- Read `show etherchannel summary` by its flags: (P) bundled, (s) suspended for a mismatch, (I) stand-alone, (D) down.

NEXT

Chapter 12 — Rapid PVST+ and the Protection Features